

CS112 – Level 1

Programming Languages 1

(*General Division – Software Engineering Program – Medical Informatics Program*)

Lecture 0

Course Introduction & Plan .. + .. a Revision on Basic Programming Concepts

Spring 2020 (Semester 2)

Week 1

Course & Lectures are based on their counterparts in the following courses:

- *Harvard University's CS50; An introduction to the intellectual enterprises of computer science and the art of programming*, Harvard School of Engineering and Applied Sciences.
- *UNSW's CS1: Higher Computing*, by Richard Buckland – The University of New South Wales.
- *MIT's 6.087 Practical Programming in C* (2010), *6.096 Introduction to C++* (2011), and *6.S096 Introduction to C and C++* (2013), MIT (Massachusetts Institute of Technology) OpenCourseWare.

Outline

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Course Details

Course Code	CS112
Course Name	Programming Languages 1 • <i>General Division</i> • <i>Software Engineering Program [SEP]</i> • <i>Medical Informatics Program [MIP]</i>
Coordinating Unit	Department of Computer Science, Faculty of Computers & Artificial Intelligence, Helwan University
Term	Semester 2 (Spring)
Level	Undergraduate - Level 1

Course Staff: Instructors (Short Bio)

(CS112 Programming Languages 1 – General Division)

Dr. Wessam El-Behaidy

- Bachelor in Computer Science *from* Helwan University, Egypt
- Masters in Computer Science *from* Helwan University, Egypt
- PhD. in Computer Science (Bio-Informatics) *from* Helwan University, Egypt

Dr. Mohammed El-Said

- Bachelor in Computer Science *from* Helwan University, Egypt
- Masters in Computer Science *from* Helwan University, Egypt
- PhD. in Computer Science (Computational Linguistics) *from* Helwan University, Egypt

Course Staff: Instructors (Short Bio)

(CS112 Programming Languages 1 – Medical Informatics Program)

Dr. Amr S. Ghoneim

- Bachelor in Computer Science *from* Helwan University, Egypt
- Masters in Computer Science *from* Helwan University, Egypt
- PhD. in Computer Science (Artificial Intelligence) *from* University of New South Wales, Australia

Dr. Milad Abu El-Khair

- Bachelor in Computer Science *from* Helwan University, Egypt
- Masters in Computer Science *from* Helwan University, Egypt
- PhD. in Computer Science (Software Engineering) *from* Helwan University, Egypt

Course Staff: Instructors (Short Bio)

(CS112 Programming Languages 1 – Software Engineering Program)

Dr. Amr S. Ghoneim

- Bachelor in Computer Science *from* Helwan University, Egypt
- Masters in Computer Science *from* Helwan University, Egypt
- PhD. in Computer Science (Artificial Intelligence) *from* University of New South Wales, Australia

Tentative Topics

Learning

Objectives

.. *of this course*

Learning Objectives of this Course

What you'll learn?

- ❖ A broad and robust understanding of programming.
- ❖ How to think algorithmically and solve programming problems efficiently.
- ❖ Concepts like abstraction, files, recursion, algorithms, sorting, searching, and data structures.
- ❖ Familiarity with the C (and C++) programming language.

The course is designed to provide more knowledge of the C/C++ language. Students will be able to develop logics which will help them to create programs/applications in C/C++. Also by learning the basic programming constructs they can easily switch over to any other language in the future.

Learning Objectives of this Course *.. Continued..*

As well, Students by the end of this course should be able to:

- ❖ Think more methodically;
- ❖ Program procedurally;
- ❖ Represent and process information;
- ❖ Solve problems efficiently;
- ❖ Recognize patterns among problems;
- ❖ Decompose problems into parts;
- ❖ Operate at multiple levels of abstraction;
- ❖ Separate design from implementation details;
- ❖ Infer from first principles how systems work;
- ❖ Assess the correctness, design, and style of code;
- ❖ Teach themselves new languages;

Topics Covered

The main contents of the course are:

- Recursive Functions
- Call by Reference versus Call by Value
- One-Dimensional Arrays
- Multidimensional Arrays
- Structs, Enums, Unions, and Type Defs
- Strings (Arrays of Characters) and String Manipulation.
- Pointers
- Search Algorithms
- Sorting Algorithms
- Handling Files (Accessing & Processing)
- Basic Data Structures

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Course Logistics

Course Logistics

- Course Materials & Announcements
- Assessment Summary (Grading Policy)
- Academic Integrity & Plagiarism

Course Materials & Announcements

All course material (lecture notes “slides”, announcements, assignments, any supplemental notes or documentation), will be made available “posted” online on weekly basis, on the webpage:

<https://www.facebook.com/amr.s.ghoneim/>

Required Resources:

The prescribed textbook for the course is:

“**C How to Program**”, by Deitel and Deitel, Pearson Education Inc., 8th edition."

Course Materials & Announcements

**Please note that the prerequisite for this course [CS112] is [CS111].
I.e., you must have successfully completed CS111 before you can enroll in CS112.**

CS111 (*Introduction to Computer Science*) provided the required foundation knowledge in order to progress into this course. **Thus, ALL the material / concepts of CS111 are included in CS112.**

The 12 lectures of the course (**CS111** – *Introduction to Computer Science* – the *Fall 2019 semester*) are available (*as single file*) at:

<https://docdro.id/VQg4YaY>

Course Materials & Announcements

For the **General Division** Students:

Dr. Mohammed El-Said will be posting announcements on that page:

<http://Fcih.tk/pl1>

Assessment Summary (Grading Policy)

The Assessment for this subject consists of six components with the following weightings (grading breakdown):

- 1) Final Written Exam: **50%**
- 2) Midterm Written Exam: **20%**
- 3) Quizzes (3 to 4 Written/Practical quizzes): **20%**
- 4) Final Practical Exam: **10%**
- 5) Attendance and Participation during the Labs. (Sections) are obligatory.
- 6) Take-home Assignments / Sheets (~5 Sheets): **0% to -10%**
(Students will **lose 2 marks for every sheet not delivered on time or found out to be copied**. The solutions should be handwritten and NOT typed/printed).

Academic Integrity & Plagiarism

- You can discuss ideas and methodology for the homework (assignments / sheets) with other students in the course, but ***you must write your solutions completely independently.***
- We will be code-checking to assess similar submissions or submissions that use code from other sources.

Previous Course
Statistics *CS111* .. & ..
Dependent Courses
*.. a list of the Courses
requiring CS112 as a
Prerequisite ..*

**The
Grading /
GPA
System
Per Course**

النقاط	التقدير	النسبة المئوية للدرجة
٤	A+	٩٠% فأكثر
٣.٧	A	٨٥% - أقل من ٩٠%
٣.٣	B +	٨٠% - أقل من ٨٥%
٣	B	٧٥% - أقل من ٨٠%
٢.٧	C +	٧٠% - أقل من ٧٥%
٢.٤	C	٦٥% - أقل من ٧٠%
٢.١	D+	٦٠% - أقل من ٦٥%
١.٨	D	٥٠% - أقل من ٦٠%
صفر	F	أقل من ٥٠%

**The
Overall
Grading /
GPA
System**

المعدل التراكمي	التقدير العام
٣.٤ فأكثر	ممتاز
٢.٨ إلى أقل من ٣.٤	جيد جداً
٢.٤ إلى أقل من ٢.٨	جيد
٢ إلى أقل من ٢.٤	مقبول
١.٤ إلى أقل من ٢	ضعيف
أقل من ١.٤	ضعيف جداً

Introduction to Computer Science (Fall 2019)

Course Statistics Average for ALL Divisions / Programs

Grade	Percentage
A+	0.7 %
A	2.2 %
B+	6.7 %
B	9.4 %
C+	10.6 %
C	15.5 %
D+	16.2 %
D	24.8 %
F	13.9 %

Super! That was first class work!!
You've almost mastered that!! ..
So, Keep up the good work :-)

Nice job! You're on the right track.
If you put in more effort & practice more
frequently, you'll get more results!
We believe you Can Do It ;-)

We appreciate your effort, & you ARE improving.
But, apparently you still have not understood some
important concepts. You need to work harder &
smarter, to practice more, & to be more engaged
with your studies. Never give up! We believe in
you, we are here for you, so make us proud! :-)

Introduction to Computer Science (CS111)

Programming Languages 1 (CS112)

Software
Engineering
1 (CS251)

Data Structures
(CS214)

Programming
Languages 2
(CS213)

Operating
Systems 1
(CS241)

Algorithms
(CS316)

Multimedia
(IT433)

Computer
Graphics (IT331)

Artificial
Intelligence
(CS361)

Software
Engineering 2
(CS352)

Graduation
Project (CS498)

Compilers (CS419)

Programming
Languages 3
(CS313)

Concepts of
Programming
Languages (CS317)

Operating Systems
2 (CS342)

Image Processing 1
(IT441)

Computer Vision
(IT444)

Computer Graphics
2 (IT332)

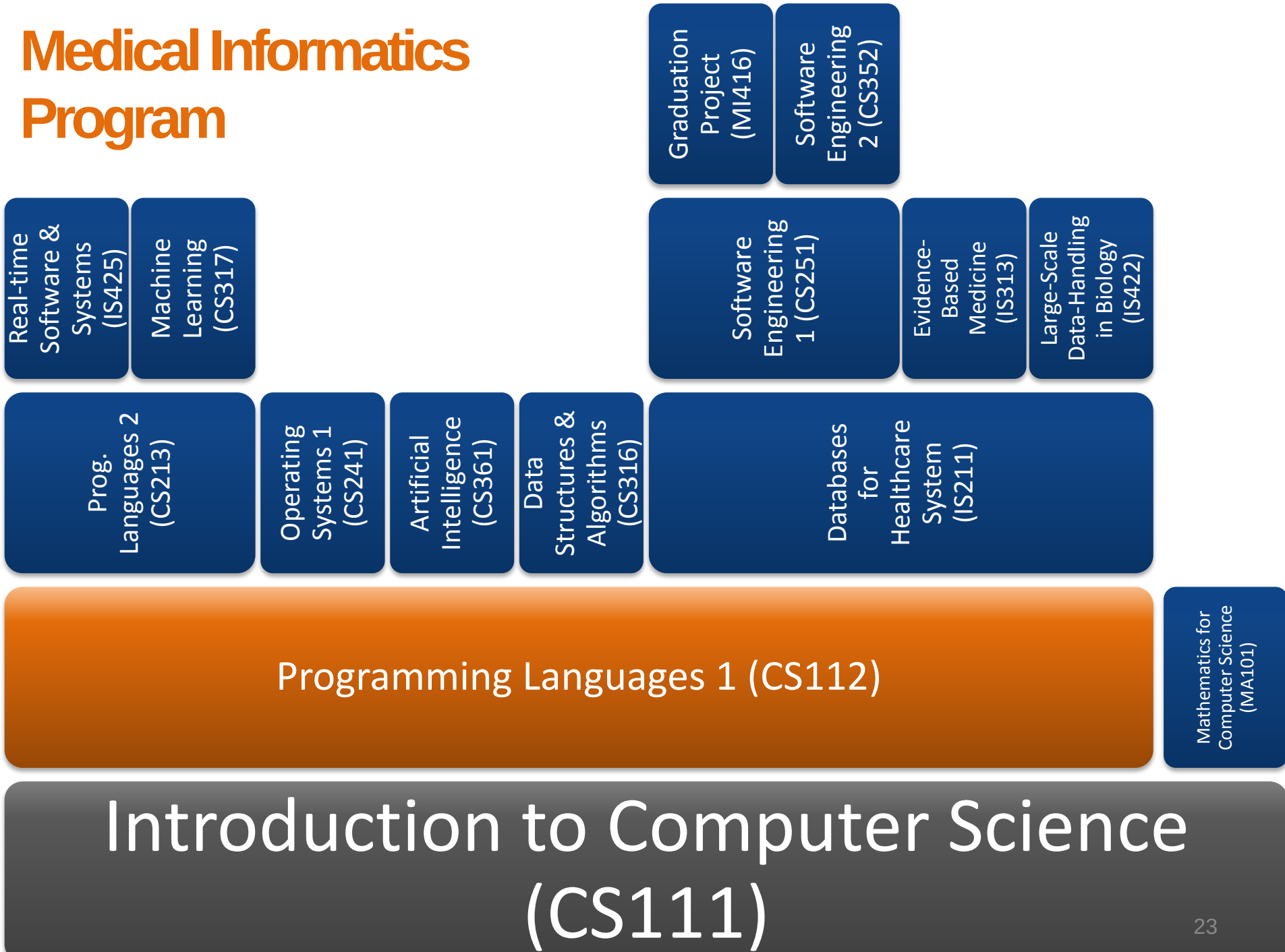
Virtual Reality
(IT431)

Intelligent
Information
Systems (IS441)

Natural Language
Processing (CS462)

General
Division

Medical Informatics Program



Software Engineering Program

Data Structures
(CS214)

Operating
Systems 1
(CS241)

Programming
Languages 2
(CS213)

Algorithms
(CS316)

Artificial
Intelligence
(CS361)

Computer
Graphics 1
(IT331)

Programming Languages 1 (CS112)

Introduction to Computer Science
(CS111)

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Academic Agenda

Week #	Starts on Sat. the:	Indicative Syllabus	Quizzes, Assignments & Exams
Week 1	Feb. 08	Course Intro. & Plan Revision on Basic Concepts	-
Week 2	Feb. 15	Recursion in Functions, Call by Reference/Value	Sheet 1 (0%)
Week 3	Feb. 22	Arrays	-
Week 4	Feb. 29	Structs, Enums, & Type Defs	Quiz 1 (5%) + Sheet 2 (0%)
Week 5	March 07	Characters/Strings	-
Week 6	March 14	Pointers [Part 1]	Quiz 2 (5%) + Sheet 3 (0%)
Week 7	March 21	Midterm Written Exams (20%) – Schedule TBA by the Faculty	
Week 8	March 28	Pointers [Part 2]	-
Week 9	April 04	Search & Sorting Algorithms	Quiz 3 (5%) + Sheet 4 (0%)
Week 10	April 11	Files Access & Processing	-
Week 11*	April 18	Basic Data Structures	Quiz 4 (5%) + Sheet 5 (0%)
Week 12	April 25	Revision	-

Public Holidays: Sham Al-Naseem (Easter Monday): April 20th .. Sinai Liberation Day: Saturday April 25th .. *Ramadan likely starts on Thursday the 23rd (week 11)

Final Practical Exams (10%) – Schedule TBA by the Faculty

Final Written Exams (50%) – Schedule TBA by the Faculty

a Revision on **Basic
Programming
Concepts
Solved Examples**

0.1: C program to print a 'Full Pyramid' Pattern.

- In this C program, we are going to learn how to print a Full pyramid using 'asterisks' and 'loops'. We are reading the total number of rows, and printing the Full pyramid accordingly.

```
Enter number of rows    10
  *
 ***
*****
*****
*****
*****
*****
*****
*****
*****
*****

-----
Process exited after 1.173 seconds with return value 0
Press any key to continue . . . _
```

```
#include<stdio.h>
```

```
int main() {
```

```
    int a, b, c, l = 1, rows;
```

```
    printf("Enter number of rows\t");
```

```
    scanf( "%d", &rows );
```

```
    for( a = 1; a <= rows; a++ )
```

```
    {
```

```
        for( b = rows; b >= a; b-- )
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        for( c = 1; c <= l; c++ )
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        l = l + 2;
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

0.2: C program to print the size of variables using the *sizeof()* built-in function.

- `sizeof()` is a built-in function in the C programming language, which is used to get the occupied size by a variable or a value. This program demonstrates the usage of the `sizeof()` function by printing the size of different types of variables.

Size of a: 1

Size of b: 4

Size of c: 4

Size of d: 8

0.2: C program to print the size of variables using the *sizeof()* built-in function.

```
/*C program to print the size of variables using sizeof() function.*/
#include <stdio.h>
int main()
{
    char a = 'A';
    int b = 120;
    float c = 123.0f;
    double d = 1222.90;

    printf("\nSize of a: %d", sizeof( a ) );
    printf("\nSize of b: %d", sizeof( b ) );
    printf("\nSize of c: %d", sizeof( c ) );
    printf("\nSize of d: %d", sizeof( d ) );
    return 0;
}
```

0.3: C program to print all uppercase and lowercase alphabets.

- In this C program, we are going to print all uppercase alphabets (from 'A' to 'Z') and lowercase alphabets (from 'a' to 'z').
- To print uppercase alphabets, we are using a for loop from 'A' to 'Z' and printing the characters, and to print lowercase alphabets, we are using another for loop from 'a' to 'z' and printing the characters.

Capital (upper) case characters:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Lower case characters:

a b c d e f g h i j k l m n o p q r s t u v w x y z

0.3: C program to print all uppercase and lowercase alphabets.

```
/*C program to print all upper case and lower case alphabets.*/  
#include <stdio.h>  
int main()  
{  
    char i;  
  
    printf("Capital (upper) case characters:\n");  
    for( i = 'A'; i <= 'Z'; i++ )  
        printf("%c ", i );  
  
    printf("\n\nLower case characters:\n");  
    for( i = 'a'; i <= 'z'; i++ )  
        printf("%c ", i );  
    return 0;  
}
```

0.4: C program to print Square, Cube, and Square Root of all the numbers from 1 to N.

- This program will print the Square, Cube, and Square Root of all the numbers from 1 to N, using a loop.
- Here, we are reading a value of N (limit) and will calculate and print the square, cube, and square root of all the numbers from 1 to N.
- To find the square; we are using $(i*i)$, the cube; we are using $(i*i*i)$, and the square root, we are using the built-in function `sqrt(i)`.
- Here, `i` is the loop counter, and `sqrt()` is a function of the `math.h`, which returns the square root of a number.

0.4: C program to print Square, Cube, and Square Root of all the numbers from 1 to N.

No	Square	Cube	Square Root
1	1	1	1.00
2	4	8	1.41
3	9	27	1.73
4	16	64	2.00
5	25	125	2.24
6	36	216	2.45
7	49	343	2.65
8	64	512	2.83
9	81	729	3.00
10	100	1000	3.16

0.4: C program to print Square, Cube, and Square Root of all the numbers from 1 to N.

```
/*C program to print the square, cube, & square root of all the numbers from
1 to N.*/
#include <stdio.h>
#include <math.h>
int main()
{
    int i, n;
    printf("Enter the value of N: "); scanf("%d", &n );
    printf("No    Square        Cube            Square Root\n", n );
    for( i = 1; i <= n; i++ )
    {
        printf("%d \t %ld \t %ld \t %.2f\n", i, ( i * i ), ( i * i * i ), sqrt( (double) i ) );
    }
    return 0;
}
```

0.5: C program to print All Leap Years from 1 to N.

- This program will read value of N and print all Leap Years from 1 to N years.
- There are two conditions for a leap year:
 - If the year is divisible by 400 (for Century years),
 - If the year is divisible by 4 and must not be divisible by 100 (for Non Century years).

0.5: C program to print All Leap Years from 1 to N.

Enter the value of N: 2000

Leap years from 1 to 2000:

4	8	12	16	20	24	28	32	36	40
44	48	52	56	60	64	68	72	76	80
84	88	92	96	104	108	112	116	120	124
128	132	136	140	144	148	152	156	160	164
168	172	176	180	184	188	192	196	204	208
...									
1820	1824	1828	1832	1836	1840	1844	1848	1852	1856
1860	1864	1868	1872	1876	1880	1884	1888	1892	1896
1904	1908	1912	1916	1920	1924	1928	1932	1936	1940
1944	1948	1952	1956	1960	1964	1968	1972	1976	1980
1984	1988	1992	1996	2000					

```
/*C program to print all leap years from 1 to N.*/  
#include <stdio.h> //function to check leap year  
int checkLeapYear( int year ) {  
    if( ( year % 400 == 0 ) || ( year % 4 == 0 && year % 100 != 0 ) )  
        return 1;  
    else return 0; }  
  
int main() {  
    int i, n;  
    printf("Enter the value of N: ");  
    scanf("%d", &n);  
    printf("Leap years from 1 to %d:\n", n);  
    for( i = 1; i <= n; i++ )  
    {  
        if( checkLeapYear( i ) )  
            printf("%d\t",i);  
    }  
    return 0;  
}
```

0.6: C program to find the HCF (Highest Common Factor) of two numbers..

- This program will read two integer numbers and print the Highest Common Factor (HCF).

Enter the first number: 100

Enter the second number: 40

HCF (Highest Common Factor) of 100, 40 is: 20

```
#include <stdio.h> /*C program to find the HCF of two numbers.*/
int findHcf( int a, int b) { //function to find HCF of two numbers
    int temp;
    if( a == 0 || b == 0 )
        return 0;
    while( b != 0 ) {
        temp = a % b;
        a = b;
        b = temp;
    }
    return a;
}

int main() {
    int a, b; int hcf;
    printf("Enter the first number: "); scanf( "%d", &a );
    printf("Enter the second number: "); scanf( "%d", &b );
    hcf = findHcf( a, b );
    printf("HCF (Highest Common Factor) of %d, %d is: %d\n", a, b, hcf );
    return 0;
}
```

Thanks! .. *Questions?*